MacBook Pro (Retina, 15-inch, Mid 2014) - Technical Specifications



Display

- Retina display: 15.4-inch (diagonal) LED-backlit display with IPS technology; 2880-by-1800 resolution at 220 pixels per inch with support for millions of colors
- Native resolution: 2880 by 1800 pixels (Retina); scaled resolutions: 1920 by 1200, 1680 by 1050, 1280 by 800, and 1024 by 640 pixels

Processor

- **2.2GHz**
  2.2GHz quad-core Intel Core i7 processor (Turbo Boost up to 3.4GHz) with 6MB shared L3 cache
  Configurable to 2.5GHz quad-core Intel Core i7 (Turbo Boost up to 3.7GHz) with 6MB shared L3 cache or 2.8GHz quad-core Intel Core i7 (Turbo Boost up to 4.0GHz) with 6MB shared L3 cache.
- **2.5GHz**
  2.5GHz quad-core Intel Core i7 processor (Turbo Boost up to 3.7GHz) with 6MB shared L3 cache
  Configurable to 2.8GHz quad-core Intel Core i7 (Turbo Boost up to 4.0GHz) with 6MB shared L3 cache.

Memory

- 16GB of 1600MHz DDR3L onboard memory

Storage[1]

- **2.2GHz - 256GB**
  256GB PCIe-based flash storage
  Configurable to 512GB or 1TB flash storage.
- **2.5GHz - 512GB**
  512GB PCIe-based flash storage
  Configurable to 1TB flash storage.

Size and Weight

- Height: 0.71 inch (1.8 cm)
- Width: 14.13 inches (35.89 cm)
- Depth: 9.73 inches (24.71 cm)
- Weight: 4.46 pounds (2.02 kg)[2]

Graphics and Video support

- Intel Iris Pro Graphics
- NVIDIA GeForce GT 750M with 2GB of GDDR5 memory and automatic graphics switching
- Dual display and video mirroring: Simultaneously supports full native resolution on the built-in display and up to 2560 by 1600 pixels on up to two external displays, both at millions of colors
- **Thunderbolt digital video output**
  - Native Mini DisplayPort output
  - DVI output using Mini DisplayPort to DVI Adapter (sold separately)
  - VGA output using Mini DisplayPort to VGA Adapter (sold separately)
  - Dual-link DVI output using Mini DisplayPort to Dual-Link DVI Adapter (sold separately)
- **HDMI video output**
  - Support for 1080p resolution at up to 60Hz
  - Support for 3840-by-2160 resolution at 30Hz
  - Support for 4096-by-2160 resolution at 24Hz

Camera

- 720p FaceTime HD camera

Connections and Expansion



- MagSafe 2 power port
- Two Thunderbolt 2 ports (up to 20 Gbps)
- Two USB 3 ports (up to 5 Gbps)
- HDMI port
- Headphone port
- SDXC card slot
- Apple Thunderbolt to FireWire Adapter (sold separately)
- Apple Thunderbolt to Gigabit Ethernet Adapter (sold separately)

Wireless

- **Wi-Fi**
  802.11ac Wi-Fi wireless networking;[3] IEEE 802.11a/b/g/n compatible
- **Bluetooth**
  Bluetooth 4.0 wireless technology

Audio

- Stereo speakers
- Dual microphones
- Headphone port
  - Support for Apple iPhone headset with remote and microphone
  - Support for audio line out (digital/analog)

Keyboard and Trackpad

Full-size backlit keyboard with 78 (U.S.) or 79 (ISO) keys, including 12 function keys and 4 arrow keys (inverted "T" arrangement) with ambient light sensor.

Multi-Touch trackpad for precise cursor control; supports inertial scrolling, pinch, rotate, swipe, three-finger swipe, four-finger swipe, tap, double-tap, and drag capabilities.

Battery and Power[4]

- Up to 8 hours wireless web
- Up to 8 hours iTunes movie playback
- Up to 30 days standby time
- Built-in 95-watt-hour lithium-polymer battery
- 85W MagSafe 2 Power Adapter with cable management system; MagSafe 2 power port

Electrical and Operating Requirements

- Line voltage: 100V to 240V AC
- Frequency: 50Hz to 60Hz
- Operating temperature: 50° to 95° F (10° to 35° C)
- Storage temperature: -13° to 113° F (-25° to 45° C)
- Relative humidity: 0% to 90% noncondensing
- Operating altitude: tested up to 10,000 feet
- Maximum storage altitude: 15,000 feet
- Maximum shipping altitude: 35,000 feet

Operating System
OS X Mavericks: Easy to use, incredibly powerful, and packed with features you'll love, OS X Mavericks is the world's most advanced desktop operating system. Learn more.

OS X Mavericks Includes

- AirDrop
- AirPlay
- Dictation
- Gatekeeper
- Notification Center

- Facebook Integration
- Twitter Integration
- Multiple Display Support
- Finder Tabs and Tags
- iCloud
- iCloud Keychain
- Dashboard
- Mission Control
- Spotlight Search
- Safari Power Saver
- App Nap
- Power Nap
- Compressed Memory
- Built-in Sharing

Built-in Apps[5]

- iPhoto
- iMovie
- GarageBand
- Pages
- Numbers
- Keynote
- Maps
- iBooks
- Safari
- Mail
- FaceTime
- Messages
- Calendar
- Contacts
- Reminders
- Time Machine
- Photo Booth
- iTunes
- Game Center
- Preview
- Notes
- Mac App Store

Limited Warranty and Service
Your MacBook Pro with Retina display comes with 90 days of free telephone support and a one-year limited warranty. Purchase the AppleCare Protection Plan to extend your service and support to three years from your computer's purchase date. Only the AppleCare Protection Plan provides you with direct telephone support from Apple technical experts and the assurance that repairs will be handled by Apple-authorized technicians using genuine Apple parts. For more information, visit Apple Support or call 800-823-2775.

In the Box

- MacBook Pro with Retina display
- MagSafe 2 Power Adapter
- AC wall plug
- Power cord

Configure to Order
Configure your MacBook Pro to order with these options, only at the Apple Online Store:

- Up to 2.8GHz quad-core Intel Core i7 (Turbo Boost up to 4.0GHz) with 6MB shared L3 cache
- Up to 1TB of flash storage[1]

MacBook Pro with Retina display and the Environment
Apple takes a complete product life cycle approach to determining our environmental impact. Learn more.

MacBook Pro with Retina display is designed with the following features to reduce its environmental impact:

- Highly recyclable aluminum enclosure
- Mercury-free LED-backlit display
- Arsenic-free display glass
- BFR-free
- PVC-free[6]

- Reduced packaging volume
- Meets ENERGY STAR 6.0 requirements
- Rated EPEAT Gold[7]

Apple and the Environment
Learn more about Apple's dedication to reducing the environmental impact of our products and process. Or read our Product Environmental Reports for detailed information on the environmental performance of every Apple product.

Recycling
Apple takes a holistic view of materials management and waste minimization. Learn more about how to recycle your Mac.

Acoustic Performance
Declared noise emission values in accordance with ECMA-109

	Sound Power Level $L_{WA,m}$ (B)	Sound Pressure Level Operator Position $L_{pA,m}$ (dB)
Idle	1.40 (K_V=0.25)	6.0
Wireless web	1.40 (K_V=0.25)	5.5

1. $L_{WA,m}$ is the mean A-weighted sound power level, rounded to the nearest 0.05 B.
2. $L_{pA,m}$ is the mean A-weighted sound pressure level measured at the operator position (rounded to the nearest 0.5 dB).
3. 1 B (bel) = 10 dB (decibel)
4. K_V is the statistical adder for computing upper-limit of A-weighted sound power level.
5. The quantity, $L_{WA,c}$ (formerly called L_{WAd}) may be computed from the sum of $L_{WA,m}$ and K_V.
6. The Wireless web test browses 25 popular websites.
7. Configuration tested: 2.5GHz quad-core Intel Core i7 processor, 16GB memory, 512GB flash storage, Intel Iris Pro & NVIDIA GeForce GT 750M graphics.

Accessories
Mac Software

- Aperture
- Logic Pro X
- Final Cut Pro X

Displays and Adapters

- Apple Thunderbolt Display
- Apple Mini DisplayPort to DVI Adapter
- Apple Mini DisplayPort to VGA Adapter
- Apple Mini DisplayPort to Dual-Link DVI Adapter
- MagSafe to MagSafe 2 Converter

AirPort and Wireless

- AirPort Express
- AirPort Extreme
- AirPort Time Capsule

Other Accessories

- Apple USB SuperDrive
- Thunderbolt to Gigabit Ethernet Adapter
- Thunderbolt to FireWire Adapter
- 60W MagSafe 2 Power Adapter
- 85W MagSafe 2 Power Adapter
- AppleCare Protection Plan

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1. 1GB = 1 billion bytes and 1TB = 1 trillion bytes; actual formatted capacity less.
2. Weight varies by configuration and manufacturing process.
3. Requires Internet access. 802.11ac is based on an IEEE draft specification.
4. Testing conducted by Apple in July 2014 using preproduction 2.5GHz quad-core Intel Core i7-based 15-inch MacBook Pro units, preproduction 2.2GHz quad-core Intel Core i7-based 15-inch MacBook Pro units, and

preproduction 2.8GHz dual-core Intel Core i5-based 13-inch MacBook Pro units. The wireless web test measures battery life by wirelessly browsing 25 popular websites with display brightness set to 12 clicks from bottom or 75%. The HD movie playback test measures battery life by playing back HD 720p content with display brightness set to 12 clicks from bottom or 75%. The standby test measures battery life by allowing a system, connected to a wireless network and signed in to an iCloud account, to enter standby mode with Safari and Mail applications launched and all system settings left at default. Battery life varies by use and configuration. See www.apple.com/batteries for more information.

5. Pages, Numbers, and Keynote are free on the Mac App Store for qualifying Mac computers purchased on or after October 1, 2013. OS X Mavericks required. Downloading apps requires an Apple ID.
6. PVC-free AC power cord available in all regions except India and South Korea.
7. MacBook Pro achieved a Gold rating from EPEAT in the U.S. and Canada.

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